

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	29 September 2026
Team ID	SWTID-2026-6203
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value

Technical Architecture:

The system is implemented on the ServiceNow platform. The application uses User Administration, custom roles, a custom Institution Details table and Access Control Lists (ACLs). When a user requests a record, ServiceNow checks the applicable ACL, verifies the required role and evaluates the Branch condition. The script then allows or denies access. Administrators retain full access.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User & Role Management	Creates the EEE test user and custom roles bb1, bb2, bb3 and bb4.	ServiceNow User Administration, Roles
2	Institution Details Table	Stores student roll number, student name, faculty name, branch, email, phone number and description.	ServiceNow Custom Table
3	Record Creation	Creates multiple records using different branch values such as ECE, EEE and CSE.	ServiceNow Forms / Lists
4	Read Access Control	Allows authorized users to view records according to the Read ACL and Branch condition.	ServiceNow Read ACL
5	Create Access Control	Controls who can create new records.	ServiceNow Create ACL
6	Write Access Control	Controls who can modify existing records.	ServiceNow Write ACL
7	Delete Access Control	Controls who can delete records.	ServiceNow Delete ACL
8	Script Control	Checks admin and custom role conditions and returns allow or deny.	ServiceNow Scripting, GlideSystem (gs)

Table Structure:

Field	Type
Student Roll Number	Auto Number
Student Name	Reference – User
Faculty Name	Reference – User
Branch	Choice – ECE, EEE, CSE
Email	String
Phone Number	String
Description	Multi String

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Technology Stack – Application Characteristics

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Role-Based Access	Access is controlled using the custom roles bb1, bb2, bb3 and bb4	ServiceNow Roles, ACL
2	Record-Level Security	ACLs protect individual records from unauthorized access.	ServiceNow ACL
3	Branch-Based Restriction	The Read ACL uses the Branch condition so EEE records are restricted to users.	ACLs, Data Conditions, Scripts
4	CRUD Access Control	Create, Read, Write and Delete operations are controlled separately	Create/Read/Write/Delete ACLs
5	Administrator Access	Administrator users retain full access to the records.	Admin Role
6	Access Verification	User impersonation is used to verify the permissions for different roles	ServiceNow Impersonation
7	List and Form Security	ACLs protect records when accessed through lists and forms.	ServiceNow Lists / Forms
8	Unauthorized Access Prevention	Users without the required role are denied access to protected records.	ACLs Security

ACL Access Flow:

User Request → Role Check → ACL Operation Check → Branch/Data Condition → Script Evaluation → Allow / Deny Access

Role and Permission Model:

Role	Permission / Purpose
bb1	Read access for the required EEE records
bb2	Create access
bb3	Write / edit access
bb4	Delete access
admin	Full access

Verification:

- User with bb1 role can view only EEE branch records.
- User without the required role cannot view the protected records.
- Admin user can view all records.
- bb1 + bb2 provides create access.
- bb1 + bb2 + bb3 provides view and edit access.
- bb1 + bb2 + bb3 + bb4 provides view, create, edit and delete access.

Project Technology:

ServiceNow platform, User Administration, Custom Tables, Roles, Access Control Lists (ACLs), Data Conditions, ServiceNow Scripting and GlideSystem (gs).

Expected Outcome:

The project demonstrates record-level security through script-controlled READ, CREATE, WRITE and DELETE ACLs, with access decisions based on user roles and record field values.